Stabilized-Algorithm Catalog Overview

Intellectual contribution of the literature extraction

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This overview summarizes the intellectual contribution of the catalog. It does *not* reproduce the catalog itself.

Purpose

The catalog was built to identify recurring structures in how data-dependent algorithms and selection procedures are stabilized—so that later theoretical work could search for reusable mechanisms rather than inventing a one-off construction. In the intellectual timeline it is recorded as a primary research product (approximately 20 reconstructed hours within the June–July effort), not a minor reading list.

What was extracted

Comparable fields used when processing sources included (where available):

- source and citation;
- algorithm or selection rule;
- selected object;
- stability notion;
- assumptions;
- stabilization mechanism;
- noise, perturbation, randomization, or regularization;
- primitive decomposition;
- proof strategy;
- composition rule;
- utility cost;
- downstream inference or generalization guarantee;
- limitations;
- relevance to the current project.

Representative entries (by mechanism)

Entries below are chosen to illustrate *distinct stabilization mechanisms*. They are grounded in repository digests, the H1 catalog note, and the paper index. They are not claimed to be exhaustive, and they are not themselves proofs of Formulation II.

Randomized selection / noisy max

Paper / method. Zrnic & Jordan, *Post-Selection Inference via Algorithmic Stability* (digest: `Formulation/research/literature/digest_zrnic_jordan_stability.md`).
Selected object. A (possibly randomized) selection map $\hat{S}$ (e.g. noisy / stable screening or Stable LASSO support).
Mechanism. Explicit randomization (Laplace / report-noisy-max style; noisy Frank–Wolfe) inducing (η, τ, ν)-stability of the selection law.
Guarantee. Stability transfers to post-selection CI validity with e^{η}-adjusted classical level (Thm. 2 in the digest).

Relevance. Closest technical template for "stabilize selection, then retain a classical inferential object"—subject to advisor judgment on baselines.

Algorithmic stability (distributional)

Paper / method. Same Zrnic–Jordan line; Bassily–Freund typical stability cited as related (digest §1).
Selected object. Randomized algorithm A outputting a selection.
Mechanism. Indistinguishability of the selection law from an oracle law on a high-probability set (Def. 2).
Guarantee. Composition / post-processing closure; transfer lemmas to post-selection coverage corrections.
Relevance. Clarifies that "stability" here is often about selection *laws*, which must be carefully related to Part I replace-one movement of $\hat{\theta}$.

Regularization / constrained randomized optimization

Paper / method. Stable LASSO / stable marginal screening constructions in Zrnic–Jordan (digest §2, design propositions).
Selected object. Model support or screened coordinates.
Mechanism. Optimization with injected noise and regularization-style control of sensitivity; utility propositions quantify selection-quality cost.
Guarantee. Explicit (η, τ, ν)-stability rates with stated assumptions.
Relevance. Concrete precedent for randomizing a constrained program—a structural cousin of randomizing problem (2) in the write-up.

Conformal risk control (fixed hyperparameter)

Paper / method. Angelopoulos et al., *Conformal Risk Control* (digest: `Formulation/research/literature/digest_angelopoulos_crc.md`).
Selected object. Threshold / risk-control parameter λ when treated as *prespecified* (or chosen by the CRC rule under exchangeability).
Mechanism. Monotone risk + exchangeability; not a post-selection stabilizer for a shared-data optimizer.
Guarantee. Fixed-parameter risk control / split-conformal coverage templates.
Relevance. Supplies the fixed-θ validity template behind write-up assumption (1); selecting $\hat{\theta}(\mathcal{D})$ on the same $\mathcal{D}$ breaks the prespecified premise.

Differential privacy / max-information style control

Paper / method. Max-divergence / (η, τ)-indistinguishability in Zrnic–Jordan Def. 1; H1 catalog note lists DP top-k / Thresholdout-style examples among MAIN catalog themes (`Formulation/context/overview/swarm/04_H1_catalog.md`; overview literature integration).
Selected object. Private or information-bounded released selection.
Mechanism. Noise calibrated to sensitivity; privacy/max-info budgets as stability parameters.
Guarantee. Distributional closeness of released outputs; often used as a bridge to inference corrections.
Relevance. Suggests reusable accounting language (η, τ, ν) for Part II transfer statements—without identifying DP with the live UQ theorem.

Optimization robustness / structured selection noise

Paper / method. Pattern notes synthesizing inverse-opt geometry with selection randomization (`Formulation/research/literature/patterns_inverse_opt_structured_noise.md`); lab CREDO/CREAM papers remain *context*, not the live problem (`Formulation/context/papers/INDEX.md`).
Selected object. Hyperparameter / decision variable from a constrained program.
Mechanism. Structured (not isotropic) noise on objectives/constraints, or soft near-argmin sampling—aimed at selection stability.
Guarantee. Currently a *design hypothesis* for randomizing write-up problem (2), not a completed theorem in this repository.
Relevance. Motivates Part I assumptions and Part II goal (iii): randomize selection; do not recalibrate $\mathcal{C}$.

Recurring patterns

Supported by the extracted literature (observations).

- Margins / gaps often control selection instability (near-ties are hard).
- Strong convexity, regularization, or explicit noise can reduce optimizer / selector sensitivity.
- Randomized selection smooths discontinuous hard decisions (e.g. $\arg\max$).
- Stability parameters can transfer into inference corrections (coverage inflation, PoSI constants, etc.).
- Utility / selection quality is typically traded against inferential validity.

Hypotheses motivating the broader framework (not established theorems here).

- Primitive decompositions may support reusable proof modules across algorithms.
- Composition of primitives requires care; certificates need not be freely interchangeable.
- Structured noise tied to the geometry of constrained optimization may be more efficient than isotropic noise for stabilizing $\hat{\theta}$.

Relation to the current formulation

The catalog helped motivate Formulation I and the later narrowing to Formulation II. It does **not** prove a universal stability framework, and operator-library results do **not** automatically imply the Part I–II UQ theorems. The immediate research target is the narrower Part I replace-one stability theorem under explicit assumptions (see `Problem_Writeup.pdf` and `Research_Update_Summary.pdf`).

Where to find the underlying material

- `Formulation/context/overview/swarm/04_H1_catalog.md` — H1 working-survey summary (≈25 MAIN examples claimed).
- `Formulation/context/overview/Research_Scope_Timeline_and_Authority.tex` — Catalog status, primitives list, literature integration.
- `Formulation/research/literature/` — Digests and pattern cards used for extraction.
- `Formulation/context/papers/INDEX.md` — Prioritized paper corpus in the repository.
- `Advisor_Packet/Intellectual_Timeline/Intellectual_Timeline.pdf` — Effort and role of the catalog in the research arc.

Missing artifact. No full structured catalog spreadsheet/database dump was found in the repository on 29 July 2026. If a separate Overleaf/database export exists outside this tree, it is not currently checked in; the packet therefore points to the surviving in-repo sources above.